experiments to determine how emitting different amounts of CO_2 affects the temperature of our planet. Instead, because we know that global climate models yield accurate predictions, we can use the models to perform "if-then" thought experiments: *If* we add a certain amount of CO_2 to the atmosphere, *then* global temperatures will rise by a certain number of degrees.

The predictions are sobering. If we continue with "business as usual," where emissions of CO_2 continue at current levels, by 2100 Earth's temperature will likely be 4°C higher than it was in 1900—an increase that would have dire effects for all of Earth's species, including our own. Moreover, even if all CO_2 emissions were to stop immediately, the temperature of our planet would continue to rise by an additional 0.6°C, making it 1.5°C higher in 2100 than it was in 1900. Earth will continue to warm long after emissions stop because it takes decades for the CO_2 already emitted to warm the oceans, and thus that long for the full effect of past emissions to be realized.

Finding Solutions to Address Climate Change

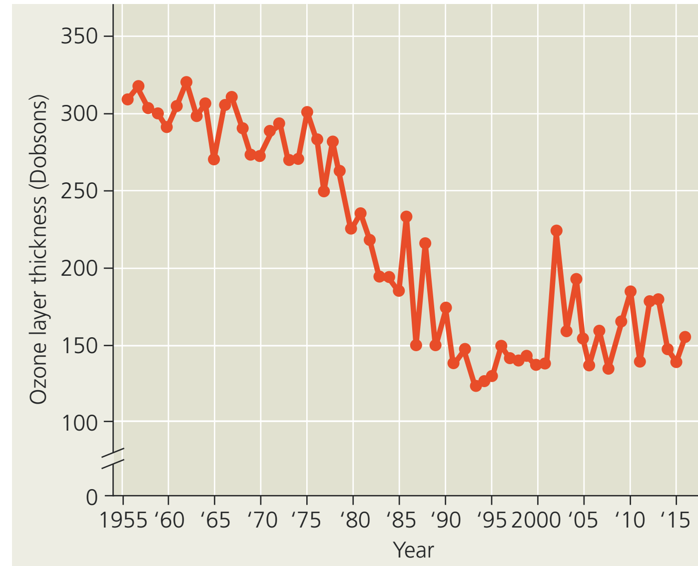
We will need many approaches to slow global warming and other aspects of climate change. Quick progress can be made by using energy more efficiently and by replacing fossil fuels with renewable solar and wind power and, more controversially, with nuclear power. Today, coal, gasoline, wood, and other organic fuels remain central to industrialized societies and cannot be burned without releasing CO_2. Stabilizing CO_2 emissions will require concerted international effort and changes in both personal lifestyles and industrial processes.

Progress toward finding solutions to address climate change was made in 2015 when all nations agreed—for the first time—to take steps to reduce CO_2 emissions and limit the extent to which global temperatures ultimately rise. This international effort, known as the Paris Agreement, has been ratified by 169 nations, including China, the United States, and all other nations that emit substantial quantities of CO_2 and other greenhouse gases. The effectiveness of the agreement was recently called into question, however, when the United States announced its intention to withdraw from the agreement by 2020. This setback highlights a potential difference between what we know and what we choose to do. An overwhelming amount of evidence indicates that climate change is real, that the rise in global temperatures since the 1950s was caused primarily by human actions, and that there will be negative consequences for human societies and all life on Earth unless we act now. What we choose to do with this information is up to us.

Depletion of Atmospheric Ozone

Like carbon dioxide and other greenhouse gases, atmospheric ozone (O_3) has also changed in concentration because of human activities. Life on Earth is protected from the damaging effects of ultraviolet (UV) radiation by a layer of ozone located in the stratosphere 17–25 km above Earth's surface. However, satellite studies of the atmosphere show that the springtime ozone

▼ Figure 56.33 Thickness of the October ozone layer over Antarctica, in units called Dobsons.

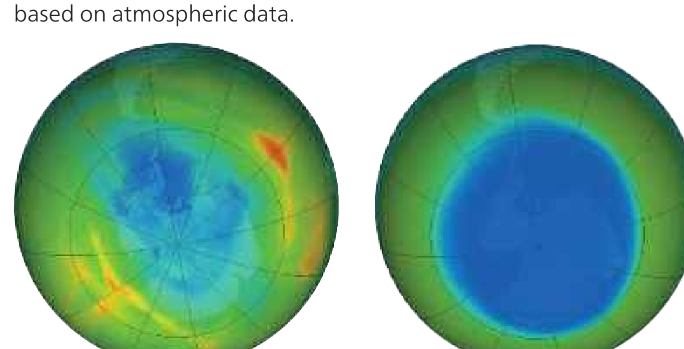


layer over Antarctica has thinned substantially since the mid-1970s **(Figure 56.33)**. The destruction of atmospheric ozone results primarily from the accumulation of chlorofluorocarbons (CFCs), chemicals once widely used in refrigeration and manufacturing. In the stratosphere, chlorine atoms released from CFCs react with ozone, reducing it to molecular O_2. Subsequent chemical reactions liberate the chlorine, allowing it to react with other ozone molecules in a catalytic chain reaction.

The thinning of the ozone layer is most apparent over Antarctica in spring, where cold, stable air allows the chain reaction to continue **(Figure 56.34)**. The magnitude of ozone depletion and the size of the ozone hole have been slightly smaller in recent years than the average for the last 20 years, but the hole still sometimes extends as far as the southernmost portions of Australia, New Zealand, and South America. At the more heavily populated middle latitudes, ozone levels have decreased 2–10% during the past 20 years.

Decreased ozone levels in the stratosphere increase the intensity of UV rays reaching Earth's surface. The consequences

▼ Figure 56.34 Erosion of Earth's ozone shield. The ozone hole over Antarctica is visible as the dark blue patch in these images based on atmospheric data.



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